Charles Benello

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Education

The University of Chicago | GPA: 3.7/4.0 Chicago, IL
Master of Science in Financial Mathematics *Expected Mar 2027*
- Coursework: C++ in Finance, Distributed Systems, Reinforcement Learning, Probability & Stochastic Processes, Options Pricing, ML in Finance

The University of Chicago | GPA: 3.75/4.0 Chicago, IL
Bachelor of Science in Mathematics and Computer Science (Systems), Minor in Korean *Jun 2025*
- Coursework: Algorithms (Honors), Data Structures, Operating Systems, Computer Architecture, Concurrency Control, Databases, Networks

Technical Skills

Languages: Python, C++, Rust, Go (Raft, gRPC), SQL, CUDA, TypeScript
Frameworks & Tools: NumPy/Pandas, PyTorch, SciPy, scikit-learn, Polars, Arrow/Parquet, Git, Docker, CMake, Linux, PostgreSQL, DuckDB, SLURM
AI Tooling: Claude Code, Codex

Experience

Software Development Engineer Intern (Incoming) Aug – Nov 2026
Amazon Web Services, Redshift Query Optimizer (planner) team: data-lake query performance *Palo Alto, CA*

Database Research Assistant Apr 2023 – Present
ChiData Research Lab, University of Chicago *Chicago, IL*
- Designed CrocSort (VLDB 2026): a skew-resilient parallel external merge sort in Rust that completes under tight memory budgets where PostgreSQL, DuckDB, and ClickHouse abort, benchmarked across skewed and uniform key distributions
- Sped up random access 3× and cut memory 40% by engineering Small String Optimization across storage structures, inlining short keys into nodes to eliminate heap allocations on the hot path
- Authored a second paper (under review) on a lightweight model that predicts optimal sort configuration on cloud and disaggregated hardware, transferring calibration across machines to within 3–5% of optimal

Machine Learning Engineer Intern Mar – Jun 2026
PayPal *Chicago, IL*
- Trained a self-supervised transaction foundation model (TabBERT-style hierarchical transformer, ∼5M params) on 14.5M unlabeled transactions, reaching AUC 0.974 / F1 0.822 on downstream fraud detection at a 0.12% positive rate
- Ran 87 ablations on a SLURM cluster across masking and pretraining objectives; per-field masking alone added +1.9% AUC and +16.8% F1

GPU Systems Researcher Jun – Sep 2025
EPFL (DIAS Lab) *Lausanne, Switzerland*
- Accelerated out-of-memory analytical queries 1.5–3× by engineering a macroblock-based GPU decompression strategy in CUDA/C++, streaming compressed macroblocks through device memory and keeping decompression on-GPU instead of round-tripping through host RAM
- Drove a new cost-based compression research direction by benchmarking GPU compression under resource-contention models, enabling adaptive decompression choices that trade decompression cost against bandwidth and contention across TPC-DS workloads

Machine Learning Engineer Intern Sept – Dec 2025
Bank of America *Chicago, IL*
- Built a portfolio-rebalancing platform that converts capital and liquidity constraints into risk-minimizing trade recommendations, stress-tested across probability-weighted macro scenarios

Software Engineering Intern Mar – Aug 2026
Alter Domus *Chicago, IL*
- Lifted document-extraction accuracy from a 50% baseline to 92–97% across document types by architecting a multi-stage LLM pipeline with schema-enforced structured output and cross-reference verification on financial filings over 100 pages
- Built an LLM-as-a-Judge evaluation framework with automated benchmarks for faithfulness, numerical accuracy, and completeness, creating the benchmark suite used to evaluate extraction quality across every supported document type

Teaching Assistant Sep 2022 – Mar 2026
University of Chicago, Computer Science & Mathematics *Chicago, IL*
- Awarded the 2026 Wayne Booth Graduate Student Teaching Prize, UChicago's top graduate teaching award
- TA'd 11 course-sections across computer science and mathematics: Head TA for Database Systems (3×), Systems Programming I–II, Intro to CS I–II, Calculus I, and Intro to Proofs in Analysis (3×)

Projects

GPU External Sort at Terabyte Scale | *CUDA/C++* | github.com/cmbenello/gpu-research 2026
- Sorted 1.08TB (9B records) in ∼5 minutes on a single H100 by streaming NVMe data directly into GPU memory with GPUDirect Storage; up to 20.1× faster than DuckDB and 34.5× faster than Polars, which fail beyond ∼200GB, making it the only tested system that completes at terabyte scale

Relational Database Engine | *Rust* | github.com/cmbenello/CrustyDB 2023
- Implemented a multi-threaded relational database from scratch: B+ tree indexing, a concurrent buffer pool with page-level locking, heap storage, and a SQL query-execution engine supporting projection, filters, joins, and aggregation

Transit Coverage Optimizer | *Python* | cmbenello.github.io/covtsp 2026
- Computed a route visiting every station on the London Underground 60 minutes faster than the standing Guinness World Record (16h45m vs. 17h45m) by solving the Covering-TSP over a 160K-node / 1.26M-edge time-expanded graph; generalized to NYC and Berlin with an interactive demo

Publications

[1] R. Otaki*, **C. Benello***, et al. "CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort." *VLDB 2026.* (*Equal contribution)
[2] **C. Benello**, R. Otaki, et al. "Getting the Most Out of External Sorting in Cloud and Disaggregated Environments." (Under review).
[3] R. Otaki, J.H. Chang, **C. Benello**, A. Elmore, G. Graefe. "Resource-Adaptive Query Execution." *CIDR 2025.*

Additional

Citizenship: US, UK, and Italy | **Interests**: Prediction markets, Rainbow Six Siege (peak top 0.1% of ∼2M ranked players), weightlifting, running (3× Chicago Marathon), climbing